

SMART INDIA HACKATHON 2026

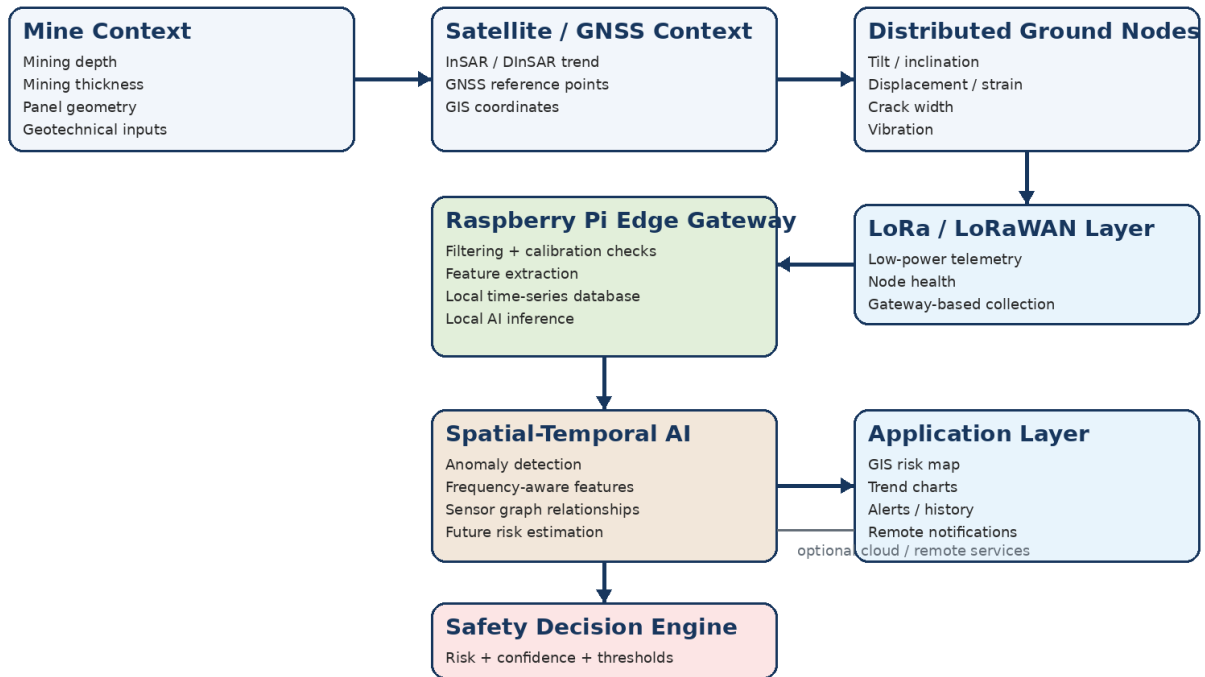
SIH26025

AI-Enabled, Low-Cost, Real-Time Mine Subsidence Monitoring, Prediction and Early Warning System

Final Technical Report

Hardware implementation + edge intelligence + spatial-temporal risk analysis

SIH26025 - Proposed System Architecture



Prepared from the problem definition and research findings discussed for the project.

1. Executive Summary

SIH26025 calls for a hardware-centric disaster-management solution capable of detecting micro-ground movements above underground coal-mine panels, estimating subsidence risk in real time, and issuing early warnings. The proposed system is a distributed Mine IoT platform in which low-cost surface sensor nodes send deformation and vibration measurements to an edge gateway. The gateway filters and validates the measurements, extracts compact features, executes AI inference locally, and drives a safety decision engine. Cloud services are optional for dashboards, history, reports, and remote notifications; loss of internet connectivity does not stop the local warning path.

The design is intentionally broader than a simple threshold alarm, but narrower than a generic all-hazards mine-management platform. The core measurements are tilt/inclination, displacement or strain, crack-width evolution, and vibration. Mining context such as seam/panel depth and thickness is incorporated as model input when available. GNSS can provide selected high-quality reference points, while InSAR/DInSAR can add wide-area deformation context and support validation. These are supporting layers rather than hard dependencies for the prototype.

Design principle

The project should be presented as Sense -> Understand -> Predict -> Warn, not simply Sensor -> Internet -> Dashboard.

2. Problem and Solution Objective

Underground coal extraction can create voids and induce movement in overlying strata. Surface subsidence is a spatial and temporal process: deformation can develop progressively, spread across neighbouring locations, and interact with local mining and geotechnical conditions. A useful early-warning platform therefore needs both continuous sensing and a way to interpret patterns rather than reacting to one isolated measurement.

2.1 Core objectives

- Continuously observe surface ground movement above mining panels using distributed low-cost sensor nodes.
- Transmit compact telemetry over a low-power wireless link such as LoRa/LoRaWAN.
- Perform filtering, feature extraction, sensor-health checks, and time-critical inference at the edge.
- Combine temporal trends, spatial relationships, mining context, and safety thresholds into a risk decision.
- Provide graded warnings and explain why a zone is considered risky.
- Continue local monitoring and alarm operation when cloud connectivity is unavailable.

2.2 Scope boundaries

The prototype is focused on subsidence monitoring and early warning. Gas, fire, water, worker localization, and broader mine-environment sensing can be added as modular future extensions, but they should not displace the subsidence pipeline in the first demonstrator. Likewise, satellite InSAR/DInSAR and GNSS are best treated as context and validation layers rather than requirements for every sensor node.

3. Research-Backed Design Insights

The research findings gathered during project development consistently point toward a multi-scale, multi-sensor architecture. Several studies examine subsidence prediction from mining and geotechnical parameters; other work demonstrates continuous strata monitoring with threshold-based alerts; remote-sensing studies show the value of DInSAR and time-series InSAR for wide-area deformation; and Mine IoT literature evaluates wireless sensor networks, LoRa/LoRaWAN, edge/cloud data handling, and reliability constraints.

Research theme	Design implication
Mining/geotechnical parameters	Mining depth, thickness and available geotechnical data should be usable as contextual features rather than ignored.
Continuous strata monitoring	Real-time monitoring and an explicit early-warning layer are central to the system.
Spatial-temporal deformation	Neighbouring sensor behaviour and movement trends should be analyzed jointly.
Time-series InSAR / DInSAR	Remote sensing can provide broad-area deformation context and validation.
GNSS-based monitoring	Selected reference points can anchor or validate low-cost distributed measurements.
Mine IoT / WSN	The system should be modular, scalable, low-power, and aware of communication quality.
Frequency-aware prediction	Slow deformation trends and fast disturbances should be separated or represented with different features.
Decision support	Outputs should include risk, confidence, zone, and recommended response level - not just raw sensor values.

Important scientific caution

Published model scores, deformation rates, and reported sensor accuracies from external studies are evidence for feasibility, not performance guarantees for this prototype. The final system must be evaluated on its own dataset and test conditions.

4. Overall System Architecture

SIH26025 - Proposed System Architecture

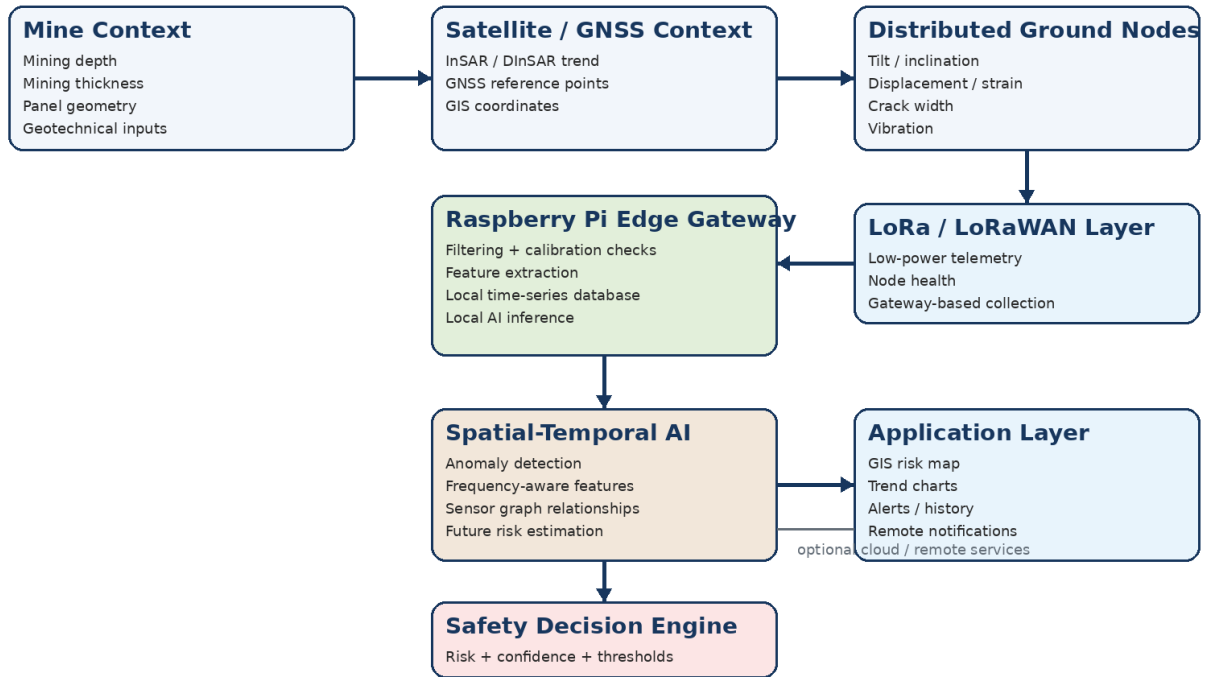


Figure 1. Proposed end-to-end architecture for SIH26025.

4.1 Architecture layers

Layer	Main elements	Purpose
Perception	Tilt, displacement/strain, crack-width, vibration	Capture physical indicators of deformation and disturbance.
Communication	LoRa or LoRaWAN, gateway	Move compact telemetry from distributed nodes to the edge.
Edge	Raspberry Pi, local database, filtering, feature extraction	Operate when offline, reduce data volume, and perform low-latency inference.
Intelligence	Anomaly detection, temporal prediction, sensor-graph analysis	Interpret current abnormality and likely future trajectory.
Safety	Thresholds + AI + confidence	Convert evidence into a graded warning decision.
Application	GIS view, trends, alerts, reports	Give mine personnel an interpretable operational picture.
Supporting context	Mining parameters, GNSS, InSAR/DInSAR	Improve contextual interpretation, mapping, and validation.

5. Hardware Implementation

5.1 Sensor node

Each surface node is a compact, low-power measurement unit. The recommended hackathon platform is an ESP32-class controller with an integrated LoRa radio board such as the Heltec WiFi LoRa 32 or a LILYGO T-Beam class device. A custom ESP32 + LoRa design is also acceptable when board cost or availability requires it.

Component	Recommended prototype	Role	Implementation notes
Microcontroller + radio	ESP32 + SX1262-class LoRa board	Local sensing, packet creation, power-state control	Use deep sleep / duty cycling where sensor dynamics allow.
Tilt / inclination	Prototype IMU such as MPU6050; field upgrade to calibrated inclinometer	Slow tilt and orientation changes	Prototype device is for demonstration; industrial deployment needs application-appropriate calibrated instrumentation.
Vibration	Accelerometer / geophone-class sensor	Short-term disturbance and vibration features	Do not equate every vibration spike with subsidence; use context and neighbouring nodes.
Displacement / strain	Strain gauge + HX711 or linear displacement prototype	Relative surface movement	Use a mechanically stable mounting arrangement and calibration routine.
Crack width	VL53L0X bracketed across a crack, or dedicated crack-width sensor	Crack opening trend	Prototype measurement depends on geometry; field deployment should use a purpose-designed crack gauge if required.
Power	Li-ion/LiFePO4 battery + solar charging	Autonomous node operation	Size from measured duty-cycle energy budget, not from a nominal panel rating.
Status indicators	LED / small display optional	Local commissioning and diagnostics	Useful for field debugging, not a required safety output.

5.2 Gateway

The gateway is the local intelligence point. A Raspberry Pi 4/5-class device with a compatible LoRa interface can receive node telemetry, maintain a local database, execute filtering and feature extraction, run the AI model, and drive local alerts. A smaller Pi-class device can be used if processing requirements remain modest.

Gateway element	Purpose
Raspberry Pi	Edge compute, local orchestration, storage, API services
LoRa interface / concentrator	Receive node telemetry
Local database	Store raw, cleaned, features and event data
UPS / battery backup (optional)	Preserve local operation during gateway power

	interruptions
Siren / beacon driver	Physical warning output through GPIO or a suitable protected interface
4G/Wi-Fi/Ethernet (optional)	Remote dashboards, notifications and synchronization

5.3 Hardware signal path

Surface node -> local sampling -> filtering / feature extraction -> LoRa packet -> gateway -> local database -> edge AI -> safety decision -> alarm / dashboard

6. Communication Architecture

LoRa is attractive for sparse, low-data-rate sensor telemetry because the node payloads are small and the deployment can span a substantial surface area. LoRaWAN can provide a standardized gateway/network architecture when a formal network-server design is desired. A simple point-to-gateway LoRa arrangement is easier for a hackathon prototype; the final report should name the architecture actually implemented rather than using LoRa, LoRaWAN, and LoRa Mesh interchangeably.

6.1 Recommended packet content

```
{ node_id, timestamp, tilt, displacement, crack_width, vibration_rms, battery, rssi, sensor_status }
```

6.2 Communication health

Every node should report enough information for the gateway to distinguish a safe measurement from missing data. A silent node is not equivalent to a safe node. The gateway should track last-seen time, battery level, received signal strength, and sensor-health flags.

6.3 ZigBee vs LoRa

Criterion	ZigBee	LoRa / LoRaWAN	Decision for SIH26025
Range	Shorter local range	Longer range	Prefer LoRa for dispersed surface telemetry.
Power	Low	Very low for small telemetry	LoRa is attractive for battery/solar nodes.
Data rate	Higher	Lower	Lower rate is acceptable because the payload is compact.
Mesh	Native mesh commonly used	LoRaWAN uses gateway architecture; mesh is a different design	Use the simplest topology that matches the prototype.
Scalability	Good for local clusters	Good for sparse distributed sensing	Gateway-centric LoRa/LoRaWAN is the preferred direction.
Internet dependency	Not inherently required	Not inherently required at the radio layer	Keep local gateway + local alarms independent of the

			internet.
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7. Signal Processing and Data Pipeline

Subsidence signals are noisy and can contain both slow deformation and fast disturbances. The preprocessing layer should preserve meaningful trends while reducing false alarms from vibration, sensor noise, mounting effects, and environmental variation.

7.1 Processing stages

1. Validate packet integrity and timestamp consistency.
2. Check sensor-health fields, battery and communication quality.
3. Apply calibration and unit conversion.
4. Filter obvious noise and outliers using a lightweight method appropriate to each signal.
5. Create sliding windows and calculate temporal features such as slope, rate of change, variance, acceleration of trend, and rolling statistics.
6. For selected signals, use frequency-aware decomposition such as VMD when the computational budget and model design justify it.
7. Create spatial features from neighbouring nodes, including agreement, correlation, cluster behaviour, and deformation spread.
8. Persist raw values, cleaned values, features, model outputs, and alert events for traceability.

7.2 Vibration feature example

Raw acceleration -> window -> RMS / peak / variance / band-energy -> compact LoRa features

The node does not need to stream a full high-rate waveform continuously to the gateway. Local feature extraction can reduce radio use and storage volume while preserving information needed for anomaly and risk models.

8. AI / ML Architecture

The recommended AI design is hybrid rather than a single monolithic model. The system separates immediate abnormality detection from future-risk estimation and combines both with engineering constraints and contextual information.

8.1 Stage A - anomaly detection

Isolation Forest, One-Class SVM, clustering/outlier methods, or another lightweight unsupervised detector can learn the normal operating envelope. The purpose is to answer: "Is this behaviour unusual now?" It should not by itself declare a mine-wide subsidence event.

8.2 Stage B - temporal prediction

A GRU/LSTM, temporal convolution model, or another lightweight sequence model can estimate the near-term trajectory of deformation-related features. The target should be defined clearly - for example,

future displacement trend or risk class - rather than claiming exact collapse time without appropriate labelled failure data.

8.3 Stage C - spatial graph intelligence

The sensor network can be represented as a graph. Nodes are sensors; edges represent proximity, panel relationships, or learned historical correlation. Graph-based methods can capture the difference between one noisy sensor and a coordinated deformation pattern across neighbouring nodes.

8.4 Context fusion

Contextual variables can include mining depth, mining thickness, panel geometry, historical movement, distance to panel boundaries, and selected geotechnical properties. Research summarized for this project indicates that mining depth can be influential in longwall subsidence prediction, while density had the least effect among the evaluated parameters in one study. The correct implementation is therefore to prioritize high-value context without pretending low-sensitivity variables are universally irrelevant.

8.5 Optional frequency-aware dual branch

Deformation signals -> frequency decomposition -> slow trend branch + disturbance branch -> spatial-temporal fusion -> risk

This design is inspired by research on VMD-based spatiotemporal prediction. It is optional for the first prototype: a simpler feature-engineered temporal model should be preferred if it yields adequate performance and much lower edge-compute cost.

9. Safety Decision Engine

Multi-Level Warning Logic

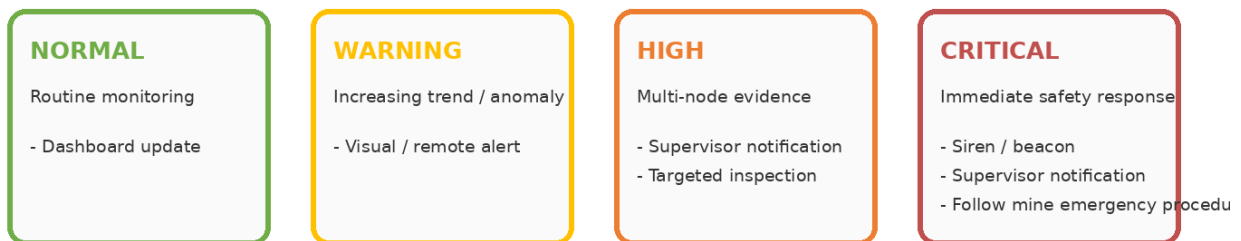


Figure 2. Recommended graded-warning logic.

The safety engine combines engineering thresholds, AI outputs, spatial agreement, temporal trend, and confidence. Thresholds are a fail-safe layer, not the entire intelligence layer.

State	Typical evidence	System response
Normal	Stable trends, healthy nodes, no meaningful anomaly	Continue monitoring; update dashboard.

Warning	Emerging anomaly or deteriorating trend	Visual warning, remote notification, closer observation.
High	Multiple neighbouring nodes agree, trend persists, context supports concern	Supervisor notification, targeted inspection, increased monitoring.
Critical	Severe threshold breach or strong multi-source evidence	Local siren/beacon and urgent notification; follow mine emergency procedure.

Safety rule

The prototype should support established mine emergency procedures. It should not claim that an experimental ML model independently commands evacuation without human and procedural safeguards.

10. Sensor Fusion and Reliability

10.1 Multi-sensor evidence

A strong warning should ideally be supported by several signals. For example, increasing displacement + increasing crack width + consistent tilt trend + agreement from neighbouring nodes is stronger evidence than a single isolated spike.

10.2 System-health state

Hazard state and system-health state should be displayed separately. The dashboard should distinguish Normal / Warning / High / Critical hazard risk from Healthy / Degraded / Offline monitoring status.

Example	Interpretation
One node abnormal; battery low; neighbours normal	Possible sensor or local disturbance; generate maintenance/inspection flag.
Several neighbours show consistent deformation and node health is good	Strong evidence of a genuine regional pattern.
Internet down but gateway and radio healthy	Local AI and physical warning remain operational.
Gateway offline	System should clearly indicate monitoring loss rather than showing the area as safe.

11. GNSS, GIS and InSAR Integration

GNSS, GIS, and InSAR/DInSAR are valuable complementary technologies, but they should not be allowed to overcomplicate the first hardware demonstrator.

Technology	Primary role	Recommended use
GNSS	Georeferenced surface displacement reference	Use at selected strategic points for validation / anchoring.
GIS	Spatial context and asset map	Use for sensor locations, panel

		boundaries, roads/buildings and risk zones.
DInSAR	Wide-area deformation measurement	Use as periodic spatial context and validation where suitable products are available.
Time-series InSAR	Longer-term deformation trend and dense spatial information	Use as an advanced monitoring layer; do not make live processing a prototype dependency.
SAR platform selection	Different sensors can perform differently for mining subsidence	Select the data source based on the actual study area, coverage and product availability; do not claim one platform is universally best.

Prototype strategy

For a hackathon demonstrator, use prerecorded or prepared InSAR/GNSS data to demonstrate fusion. Keep the core warning path fully functional with the physical sensor network and Raspberry Pi even without satellite connectivity.

12. Dashboard and User Interface

The dashboard should turn sensor streams into an operational picture. A map-first interface is preferable to a page dominated by raw telemetry.

Screen	Key elements
Mine overview	Mine zones, panel map, current risk, system-health summary
Zone detail	Tilt/displacement/crack/vibration trends, neighbouring-node agreement, risk and confidence
Alerts	Time, zone, severity, evidence, acknowledgement state, recommended action
Node health	Battery, RSSI, last-seen time, sensor-health flag, calibration state
History / reports	Trend plots, event replay, model performance, exported records

13. Recommended Prototype Bill of Materials

Item	Qty. for demo	Approx. purpose	Priority
ESP32 + LoRa development board	4-8	Sensor nodes	Core
MPU6050-class IMU / accelerometer	4-8	Tilt + vibration prototype	Core

Strain gauge + HX711 or displacement mechanism	2-4	Displacement demonstration	Core
VL53L0X or dedicated crack-width mechanism	2-4	Crack-width demonstration	Core
Raspberry Pi 4/5-class gateway	1	Edge processing + local storage	Core
LoRa gateway/concentrator interface	1	Receive node packets	Core
Battery + solar charging prototype	4-8	Autonomous nodes	Core
Siren / beacon + driver	1	Physical warning	Core
GNSS reference module	1 optional	Validation/reference	Optional
Gas/temp/humidity/water modules	As needed	Future mine-safety extension	Optional
4G modem / Wi-Fi	1 optional	Remote dashboard / notifications	Optional

14. Power and Deployment Strategy

Battery life should be designed from a measured energy budget. The node can wake, sample, compute compact features, transmit, and sleep. Sampling and transmission periods should be selected based on the dynamics of the parameter being observed; not every signal needs the same rate.

Daily energy = sensor + MCU + radio + storage/charging losses
 Solar harvest >= average daily consumption + reserve margin

For field deployment, environmental sealing, cable strain relief, mounting stability, calibration, lightning/surge protection, and battery safety become as important as the electronics themselves. Hobby-grade prototype sensors are suitable for demonstrating the concept but are not a substitute for mine-rated instrumentation in a production installation.

15. Software Stack

Layer	Suggested technology	Purpose
Node firmware	Arduino/C++ or ESP-IDF	Sensor acquisition, filtering, LoRa packetization, power management
Gateway	Python + system services	Ingestion, preprocessing, orchestration
Local database	InfluxDB / TimescaleDB / SQLite for small demo	Time-series storage
ML	Python, scikit-learn, PyTorch/TensorFlow as appropriate	Anomaly detection and prediction
Edge model	ONNX Runtime / TensorFlow Lite as appropriate	Low-overhead inference
API	FastAPI or lightweight REST service	Expose data and model outputs

Dashboard	React + map/GIS library or Grafana for rapid demo	Visualization and operator interface
Notifications	Telegram / SMS / email gateway as optional service	Remote alerts

16. AI Training and Dataset Strategy

The most difficult practical issue is obtaining labelled subsidence-failure data. The project should therefore separate three dataset types.

Dataset type	Use
Real sensor data	Calibration, noise characterization, validation and final testing where available.
Historical / remote-sensing data	Longer-term spatial deformation patterns and contextual features.
Synthetic / simulated sequences	Prototype model development, edge-case generation and demonstration before sufficient real labels exist.

Evaluation rule

Train/validation/test splits must respect time and location. Randomly mixing neighbouring time windows from the same event can make performance appear better than it really is. Final evaluation should test unseen periods, nodes or zones where possible.

17. Novelty and Differentiation

The project should not claim novelty for commodity components or generic IoT features. ESP32, LoRa, cloud dashboards, threshold alerts, and generic LSTM use are established patterns. The stronger differentiation is the way the components are combined around the subsidence problem.

Potential contribution	Why it matters
Spatial-temporal subsidence intelligence	Treats movement as a coordinated pattern across space and time rather than a single-node threshold event.
Frequency-aware edge features	Separates slow deformation trends from fast disturbances when useful.
Context-aware risk	Uses mining/panel/geotechnical context together with live measurements.
Risk + confidence output	Communicates not only severity but also strength of supporting evidence.
Sensor-health aware monitoring	Distinguishes "safe" from "not currently observed" and reduces false confidence from failed nodes.
Multi-scale integration	Allows ground IoT, GNSS, GIS and InSAR to contribute complementary evidence.

Offline-first safety path	Local gateway inference and alarms continue when cloud services are unavailable.
Modular deployment	New sensor types or communication layers can be added without redesigning the entire platform.

18. Implementation Roadmap

Phase	Deliverables
Phase 1 - Prototype node	ESP32 + LoRa + tilt/vibration; reliable packet transmission; basic battery telemetry.
Phase 2 - Ground deformation	Add displacement and crack-width mechanism; calibration and filtering.
Phase 3 - Gateway	Raspberry Pi ingestion, local database, node-health monitoring, local alarm.
Phase 4 - Baseline intelligence	Threshold engine + anomaly detection + trend features.
Phase 5 - Spatial-temporal AI	Neighbour correlation, graph representation, temporal prediction.
Phase 6 - Context + validation	Add mining context, prepared GNSS/InSAR inputs, GIS map.
Phase 7 - Demo hardening	Fault injection, communication loss tests, sensor disconnect tests, repeatable evaluation.
Phase 8 - SIH presentation	Live sensor event -> edge AI -> risk level -> physical alarm -> dashboard evidence.

19. Testing and Evaluation Plan

Test category	Example metric / test
Sensor accuracy	Compare prototype sensor response with a reference measurement under controlled displacement/tilt.
Communication	Packet delivery ratio, latency, RSSI, behaviour under partial obstruction.
Power	Average current, daily energy, estimated autonomy under actual duty cycle.
Edge AI	Inference latency, CPU/RAM use, offline operation.
Prediction	MAE/RMSE for continuous targets or precision/recall/F1 for event classification, using time-aware splits.
Early warning	Lead time between detectable precursor and configured warning event, with false-alarm rate.
Reliability	Node loss, gateway restart, sensor freeze, time-sync errors, packet corruption.

Usability	Time to understand alert cause and locate the affected zone on the dashboard.
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20. Cost-Control Strategy

The low-cost objective should come from the architecture rather than a single cheap component. The main levers are low-cost microcontrollers, small telemetry packets, low-power duty cycling, reduced cabling, modular sensor selection, edge feature extraction, and gateway-centric processing. A final bill of materials should be quoted using the actual suppliers available to the team and should include enclosures, connectors, mounting hardware, solar charging, batteries, and spares rather than only the electronic boards.

21. Key Risks and Mitigations

Risk	Mitigation
False alarms from machinery / traffic	Use vibration features, context, temporal persistence, neighbour agreement and calibration.
Sensor drift / poor mounting	Rigid installation, calibration, reference checks and node-health monitoring.
Wireless packet loss	Buffer at node, sequence numbers, retries where appropriate, last-seen monitoring, gateway placement studies.
Internet failure	Keep data storage, AI inference and physical alarm local to the gateway.
Insufficient labelled failure data	Use anomaly detection, simulation for development, historical/remote-sensing context and careful validation.
Overly complex AI	Start with lightweight models; only add VMD/graph/deep models when measured benefit justifies edge cost.
Unsafe field interpretation	Use engineering review and established mine procedures; prototype warnings are decision support, not certification.
Scope creep	Keep gas, worker tracking, and other hazards modular and secondary to subsidence.

22. Recommended Final SIH Demonstration

The most convincing live demonstration is a controlled deformation event rather than a generic dashboard tour. A sensor node should experience a known change, transmit the measurement, and show how the edge system responds.

9. Create a controlled physical tilt or crack-width change on one or more prototype nodes.
10. Show the LoRa packet arriving at the Raspberry Pi and the local node-health status.
11. Show filtering and feature extraction turning raw readings into a trend.

12. Introduce a second neighbouring node so the audience can see spatial agreement rather than a single-node trigger.
13. Display the AI risk and confidence with the evidence that produced them.
14. Trigger the appropriate physical alarm level and dashboard event.
15. Demonstrate an internet disconnect while the local gateway continues to monitor and alarm.

Judge-facing message

The differentiator is not that the system can read sensors. The differentiator is that it can turn distributed, noisy, multi-source ground observations into a local, explainable, context-aware early-warning decision.

23. Conclusion

The proposed SIH26025 solution is a layered, low-cost Mine IoT system built around continuous surface deformation monitoring and edge intelligence. Its core hardware consists of distributed ESP32-class LoRa sensor nodes measuring tilt, vibration, displacement, and crack evolution, plus a Raspberry Pi gateway for local storage, feature extraction, AI inference, and physical alarms. The intelligence layer combines anomaly detection, temporal trend analysis, spatial sensor relationships, and mining context, while a safety decision engine adds engineering thresholds and confidence-aware response levels.

GNSS, GIS, and InSAR/DInSAR improve geospatial context and validation, but the core warning path is intentionally capable of operating without them and without cloud connectivity. This keeps the prototype practical while leaving a clear path toward a larger multi-scale deployment. The result is not simply an IoT monitoring dashboard; it is a proposed decision-support and early-warning architecture designed around the specific characteristics and constraints of underground coal-mine subsidence.

24. References and Research Basis

16. Problem statement context: SIH26025 - AI-enabled, low-cost, real-time mine subsidence monitoring, prediction and early warning system (as supplied in the project discussion).
17. BlinknBuild-hosted SIH 2026 problem statement catalogue:
https://www.blinknbuild.in/Assets/SIH_2026_All_226_Problem_Statements_Master_Catalogue.pdf
18. SIH26025 problem summary: <https://zaidsayyed.in/tools/sih-problem-statements/sih26025>
19. SIH 2026 theme/problem summary:
<https://www.mtsv4.in/sih-2026-themes/disaster-management/sih26025>
20. Research finding 1: longwall coal-mine subsidence prediction using 11 mining/geotechnical parameters and optimized GEP; summarized performance included a reported correlation coefficient near 0.96 for the GEP + ABC combination and sensitivity findings for mining depth and density.
21. Research finding 2: integrated strata-management system using continuous monitoring, mine-specific thresholds, AI analysis, and alarms for advance roof/strata failure warning.
22. Research finding 3: PIV-based physical modelling of coal-mining strata movement; mining thickness and seam depth were examined in relation to deformation and subsidence.
23. Research findings 4-5: time-series InSAR and DInSAR approaches for mining-related surface deformation, including comparison with ground observations.

24. Research findings 6-9: wireless sensor networks and IoT-based mine safety monitoring using environmental sensors, LoRa/ZigBee or Wi-Fi links, threshold alerts and automated response concepts.
25. Research finding 10: integrated mine safety platform using cluster/outlier detection, spatiotemporal analysis, localization, IoT and cloud components.
26. Research finding 12: VMD-based dual-branch spatiotemporal graph modelling for nonstationary mining time series; used here as methodological inspiration for frequency-aware spatial-temporal prediction.
27. Research finding 14-15: Mine IoT reviews covering three-layer architectures, sensors, network topologies, LoRaWAN, deployment constraints, battery life, communication quality, and data management.
28. Research finding 16: automated coal-mining subsidence monitoring architecture using GNSS CORS, GIS, communication, data processing, prediction and early warning.
29. Research finding 17: comparison of ALOS-2, Sentinel-1 and Radarsat-2 InSAR for longwall mining subsidence monitoring; used here as evidence that SAR-source suitability should be evaluated for the specific study context.
30. Research finding 18: TOPSIS-based technology selection; used here as a general rationale for multi-criteria selection rather than as a universal ranking of Cloud over IoT for mining.

Appendix A - Minimal End-to-End Data Model

```
Node telemetry
- node_id
- timestamp
- latitude / longitude (or local grid coordinates)
- tilt_x / tilt_y
- displacement
- crack_width
- vibration_rms / peak / band_energy
- battery
- rssi
- sensor_health
```

```
Derived features
- slope / trend
- rolling mean / variance
- acceleration of trend
- neighbour agreement / graph features
- anomaly_score
```

```
Decision output
- risk_level
- risk_score
- confidence
- affected_zone
- evidence
- alert_state
```

Appendix B - Core Design Rules

- Do not equate one sensor spike with mine-wide subsidence.

- Do not treat a missing sensor as a safe measurement.
- Do not make cloud connectivity a prerequisite for local warning.
- Do not claim exact failure-time prediction without appropriate labelled data.
- Do not present prototype hobby sensors as mine-certified instruments.
- Do not use published model accuracy as the accuracy of the proposed system.
- Do keep the SIH prototype focused on subsidence and early warning.